

MANSI BHADRESH GOPANI

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EDUCATION

Syracuse University, School of Information Studies, Syracuse, NY

August 2023 - May 2025

Master of Science | Applied Data Science

Coursework: Applied Machine Learning, Big Data Analytics, Natural Language Processing, Database Management Systems, Responsible AI, Statistical Analysis, Business Analytics.

EXPERIENCES

Data Analyst (Academic), DJSCE, India

January 2022 - June 2023

- Collected 2,500 daily stock price records from **Yahoo Finance API** implemented Min-Max scaling and handled outliers using **Outlier Winsorization**, which improved data quality by **12%**.
- Engineered a time series forecasting model in Python using **Auto-ARIMA**, moving averages, and trend indicators, reducing **RMSE by 17%** through statistical tuning and feature optimization.
- Benchmarked ARIMA model against Prophet and LSTM models to forecast stock prices, using **Statsmodels and TensorFlow** then validated accuracy with RMSE and AIC metrics.
- Presented forecasting results to financial analysts, highlighted limitations of ARIMA for stock predictions, and recommended AI-driven approaches like **LSTM and Transformers** for improved portfolio risk management.

Data Science Intern, Gustovalley Technovations, India

August 2021 - November 2021

- Constructed a high-performance **ETL pipeline** leveraging **PostgreSQL, T-SQL, Spark, and Airflow** to ingest 33K+ health records, boosting processing speed by 40% through optimized queries and indexing.
- Extracted 100K+ features and built predictive models (**XGBoost, Random Forest**) in PySpark, achieving 74.2% R^2 , with **Snowflake and BigQuery** powering scalable cloud-based training.
- Built KPI-focused **Tableau dashboards** with geospatial insights, enabling policymakers to target uninsured hotspots and improve outreach effectiveness by **32%**, directly shaping healthcare strategies.

PROJECTS

Smart Energy Analytics: Predicting & Reducing Power Demand in SC

- Led a data-driven project for ESC (South Carolina's leading energy provider) to forecast **energy consumption** and prevent grid failures due to rising summer demand.
- Developed a predictive model using **Scikit-learn** and XGBoost; enhanced **SQL** performance by 25% using **INDEX, PARTITION BY, and COALESCE**, and reduced features from 3,171 to 93 using correlation analysis and RFE.
- Decreased RMSE by 9% and developed an **R Shiny dashboard** featuring time-series forecasting, geospatial heatmaps, and dynamic filtering, leading to **28%** projected energy savings (1.2M kWh/month) and better grid efficiency.

NLP Pipeline Development for Sentiment Analysis on Twitter Data

- Implemented a sentiment analysis model on **30K+ tweets**, tackling imbalanced data and noisy text for sentiment classification.
- Created an NLP pipeline using **NLTK**, extracting 171 features with **TweetTokenizer, SentiWordNet, POS tags**, and emoticons, and trained models with hyperparameter tuning via **RandomizedSearchCV**.
- Developed a sentiment classifier to detect opinion polarity, achieving 62% accuracy and boosting **F1-score** by refining features and resolving class imbalance with **BERT** embeddings and **SMOTE**.

AI-Driven Cold Chain Risk Intelligence on AWS

- Orchestrated an **end-to-end** cold chain risk analytics pipeline operating **Apache Airflow**, automating ingestion of sensor and weather data **into AWS S3** and **MongoDB Atlas**, with **CI/CD** integration via **GitHub Actions**.
- Trained and deployed Prophet and LSTM models on **AWS SageMaker** to estimate temperature excursions and delivery delays with 91% accuracy, enabling early intervention alerts across pharma shipments.
- Built a real-time BI solution using **Streamlit** and **Power BI** embedded on AWS for live geo-tracking, anomaly detection, and prescriptive insights, reducing simulated spoilage risk by **22%**.

TECHNICAL SKILLS

- Programming and ML:** Python (Numpy, Pandas, TensorFlow, Pytorch, Scikit-learn), R, SQL (MS SQL, T-SQL, PostgreSQL), GIT, VBA, SAS, DAX, A/B Testing, NLP, BERT, GPT-3.
- Data Engineering:** ETL/ELT, Data Lake, Data Warehousing, Data Wrangling, Scala, Hadoop, Spark, Kafka, Airflow, Azure Databricks, Docker.
- Databases & Cloud:** SQL Server, BigQuery, Redshift, Snowflake, MongoDB, Oracle, Firebase, AWS, GCP.
- Visualization & BI Tools:** Tableau, PowerBI, Google Data Studio, Plotly, QlikView, SSIS.
- Project Tools & Soft Skills:** MS Excel, Agile, Scrum, JIRA, Kanban, CRM, Project Management, SAP, PowerApps, Root Cause Analysis, Waterfall.